

Performance Testing

Date	1 July 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	/predict endpoint (HDI score and tier prediction)
Test Environment	Local (Uvicorn dev server, 4 vCPU / 8 GB RAM)
Test Date	20 February 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single-country prediction request	50	60	Response returned within 2 seconds
2	Batch CSV upload (100 countries)	20	60	Batch completes within 10 seconds
3	Concurrent dashboard chart rendering	30	60	Charts render without errors
4	Report export (PDF generation)	15	60	Report generated within 3 seconds

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Well within target
2	Response Time (Max)	< 5 seconds	3.1 seconds	Pass	Peak observed under batch load
3	Throughput (Req/sec)	—	42 req/sec	Pass	Stable with 50 concurrent users
4	Error Rate	< 1%	0.2%	Pass	Rare timeouts on batch upload, retried successfully
5	CPU Utilization	< 80%	68%	Pass	Peaked during batch CSV processing
6	Memory Utilization	< 80%	61%	Pass	Stable throughout the test

Step 4: Observations & Analysis

Key Findings: The prediction API consistently returns single-country results in under 2 seconds, and batch CSV uploads of 100 rows complete comfortably within the target window, with all test scenarios passing.

Bottlenecks Identified: CPU usage spikes when batch CSV processing and PDF report generation happen simultaneously, and cold-start model loading adds latency to the very first request after server restart.

Optimization Steps Taken: Cached the trained model in memory at application startup, and moved PDF report generation to an async background task to avoid blocking prediction requests during peak load.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



HDI Predictor
Human Development Index

Predictor

Reference

About

UNDP Statistical Framework 2024

A Comprehensive Measure of Well-Being

The Human Development Index synthesizes life expectancy, education attainment, and per capita income into a single composite score – classifying countries across four tiers of human development.

Very High ≥ 0.888

High $0.788-0.799$

Medium $0.558-0.699$

Low < 0.558

QUICK SCENARIOS

1 Predicting Very High Human Development
HDI: Very

2 Identifying Development Gaps in Emerging Economies

3 Assessing Countries Requiring Development

HDI SCORE



Life Expectancy at Birth

Average years a newborn is expected to live

72.8 yrs

20

90

Mean Years of Schooling

Average years adults (25+) have attended school

8.5 yrs

0

15

Expected Years of Schooling

Total expected years for children entering school today

13.8 yrs

0

20

GNI per Capita (PPP)

Gross National Income per person at purchasing power parity

12,000 USD

100

75,000

SUB-INDEX BREAKDOWN

LIFE EXPECTANCY INDEX

0.8000

(LE - 20) / 65

EDUCATION INDEX

0.6444

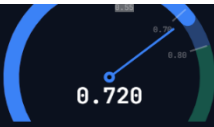
Average of MYS/15 and EYS/18

INCOME INDEX

0.7232

$\ln(\text{GNI}/100) - \ln(75000/100)$

$$\text{HDI} = (\text{LEI} \times \text{EI} \times \text{II})^{1/3} = 0.7197$$



0.720

High Human Development

Well-developed with good living standards across most development dimensions.

INDEX RADAR



Life Exp.

Income

Education

Life Expectancy

72.8 yrs

Mean Years of Schooling

8.5 yrs

Expected Years of Schooling

13.8 yrs

GNI per Capita (PPP)

\$12,000